

Arnab Bhattacharjee

Lecturer, Department of CSE, United International University, Dhaka, Bangladesh

✉ arnabbndc@gmail.com 📞 WhatsApp 🌐 arnabbndc.github.io 🔗 [LinkedIn](#)

Research Interests

Human-Computer Interaction (HCI), Agentic AI, Computer Science Education (CSEd), Social Computing, Systems

Education

Bangladesh University of Engineering and Technology (BUET)

Feb 2020 – Mar 2025

B.Sc. (Honours) in Computer Science and Engineering

CGPA: 3.88 / 4.00

Test Scores

IELTS Academic (British Council, computer-delivered): 7.5 overall (Speaking 7.0)

Dec 2025

Research Experience

Human-AI Co-Creation for Technical Lecture Content

Jan 2026 – Sep 2026

HCI, Agentic AI, CSEd | Supervisor: Dr. Novia Nurain, BUET

- Investigated how computer science educators author technical lecture content with generative AI tools.
- In a formative study, conducted semi-structured interviews with 12 university educators and analyzed them using reflexive thematic analysis to identify their challenges and derive design goals.
- Guided by these goals, built an agentic pipeline that generates technical lecture content under the educator's direction.
- Conducted a qualitative user evaluation through two focus group discussions (FGDs) with 12 educators.
- Manuscript under review at ACM CHI 2027. Co-first author.

Language Effects on LLM Web Agents

Jul 2026 – Sep 2026

Pilot research project, extension of undergraduate thesis | Supervisor: Dr. Rifat Shahriyar, BUET

- Studied whether LLM web agents, which complete tasks on websites for users, work as well in languages other than English. Translated the same web pages and tasks into seven widely spoken languages, tested an agent on them and on real websites in those languages, and found that it performs nearly as well as in English but needs considerably more tokens for non-English pages.
- Manuscript: *Measuring Multilingual Performance of HTML-based Web Agents*. Submitted, under review. Co-first author.

Evaluating the Multilingual Capabilities of LLM-based Web UI Agent

Nov 2023 – Mar 2025

Undergraduate Thesis | Supervisor: Dr. Rifat Shahriyar, BUET

- Explored AI agents that complete multi-step tasks on websites for users who give instructions in languages other than English. Fine-tuned `mT5-base` on an extended Mind2Web dataset, benchmarked the zero-shot performance of agents built on Gemini-2.0-Flash, GPT-4o-mini, and DeepSeek-V3, and built a VLM-based automated evaluator. [Extended abstract](#).
- Poster: *How well do LLM Web Agents work for non-English Language?* LLM Agent Workshop, Carnegie Mellon University, Pittsburgh, PA, USA. May 2, 2024. [PDF](#).

Academic and Professional Experience

United International University (UIU)

Jul 2025 – Present

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

- Taught Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Discrete Mathematics, Digital Logic Design, Network Security, and Engineering Ethics. Coordinator of Microprocessor and Microcontroller.
- Trained in Outcome-Based Education (OBE), the framework of UIU CSE's Washington Accord accreditation, applied in designing course curricula and lesson plans.

Samsung Research and Development Institute, Bangladesh

May 2025 – Jul 2025

Software Engineer

Dhaka, Bangladesh

- Developed an Android application (Kotlin, Android Studio) for an intelligent pen that writes on any surface, with the strokes appearing on the Android device automatically.

Projects

- Project Tokkhok** | *React, FastAPI, PostgreSQL, Supabase, RAG* | [Demo](#) | [GitHub](#) 2025
- AI-assisted learning platform for Bengali-speaking students: chat with an uploaded knowledge base (retrieval over a vector database) in Bangla, Banglish (Bangla typed in English letters), or audio, and a smart note editor that converts Banglish notes to proper Bangla.
- Buldr-Bangladesh** | *React, Spring Boot, PostgreSQL, Tailwind CSS, Firebase Auth* | [GitHub](#) 2025
- Full-stack platform for engineers to learn (AI chatbot, smart notes, summarizing and paraphrasing tools), showcase projects, sell projects through a marketplace with order tracking, and get auto-generated portfolios.
- Machine Learning Assisted Gene Expression Restoration** | *Machine Learning, Bioinformatics* | [GitHub](#) 2024
- Explored nonlinear models for identifying the transcription factors that restore gene expression from a source cell state to a target state, extending the linear GAGER approach. Nonlinear models often reduced expression differences using fewer transcription factors across several scRNA-seq datasets.
- Deepfake Image Detector Robust Against Adversarial Attacks** | *PyTorch, ViT, YOLOv11* | [GitHub](#) 2025
- Detector for real and AI-generated images (CIFAKE). Feature dropping and augmentation reached up to 80% accuracy on adversarial data, and custom adversarial noise and dropout layers improved robustness.
- Improving the xv6 Operating System** | *C, xv6 (RISC-V)* | [GitHub](#) 2023
- Added new system calls, a multilevel feedback queue (MLFQ) scheduler, inter-process communication, and threading support to the xv6 operating system.
- Mini C Compiler** | *C, Flex, Bison, x86 Assembly* | [GitHub](#) 2023
- Built a compiler front end for a subset of C with a symbol table, lexical and semantic analysis, and intermediate code generation for x86.
- TechWorld** | *Node.js, Express.js, PostgreSQL, Oracle PL/SQL* | [GitHub](#) 2022
- E-commerce website for PCs with complex SQL and PL/SQL operations. Core feature: PC-BUILDER for assembling a custom PC from compatible parts.

Skills

AI and LLMs: Agentic LLM pipelines, RAG, LLM-as-judge evaluation, PyTorch, fine-tuning transformer models
HCI Methods: Semi-structured interviews, reflexive thematic analysis, focus group discussions, formative studies

Honors and Awards

- Dean's List Scholarship, BUET (all four years)
- Champion, ITVerse 2023 "TechTales" Project Showcase Competition, University of Dhaka 2023
- Finalist, BUET CSE Fest Capture the Flag (CTF) Competition 2023
- Board General Scholarship, Secondary School Certificate (SSC) Examination 2017
- Talent Pool Scholarship, Junior School Certificate (JSC) Examination 2014
- Talent Pool Scholarship, Primary School Certificate (PSC) Examination 2011
- Competitive programming: 600+ problems solved, max rating 1519 on [Codeforces](#)

References

Available upon request.